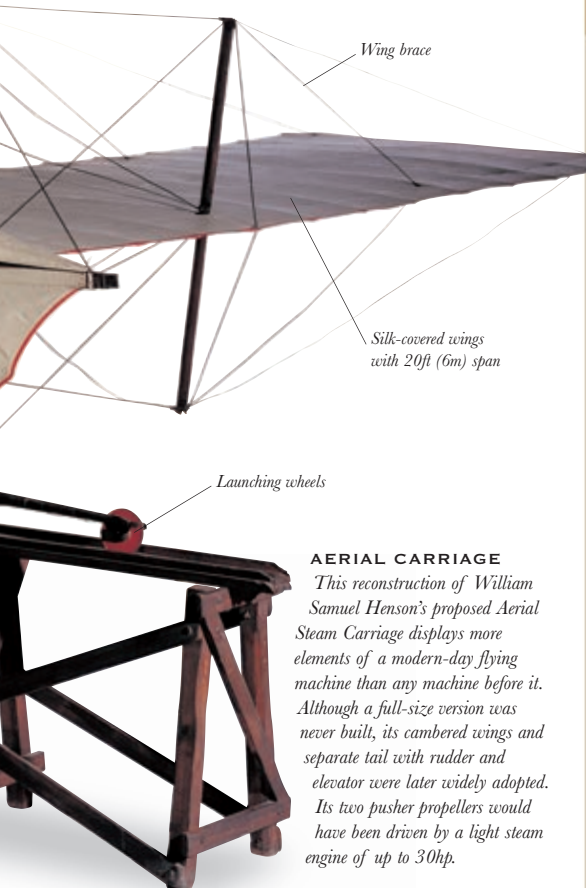




AERIAL CARRIAGE

This reconstruction of William Samuel Henson's proposed Aerial Steam Carriage displays more elements of a modern-day flying machine than any machine before it. Although a full-size version was never built, its cambered wings and separate tail with rudder and elevator were later widely adopted. Its two pusher propellers would have been driven by a light steam engine of up to 30hp.

as steam trains transformed journey times by land. At sea, steamships were a growing threat to the dominance of sail. Jumping on the bandwagon, in 1843 ambitious English inventor William Samuel Henson patented an Aerial Steam Carriage "for conveying letters, goods, and passengers from place to place."

Steam-driven airline

Basing his ideas on Cayley's published research, Henson imagined a monoplane with a cambered wing for extra lift, a rudder and tailplane for control, and two six-bladed pusher propellers. This contraption was to be powered by a 30hp steam engine in the fuselage. Henson's grandiose plans for an Aerial Steam Transit Company momentarily attracted the interest of investors, the proposal for passenger flights spanning the globe rendered credible by fanciful illustrations of the Steam Carriage soaring over exotic locations. But doubt and ridicule soon followed. Although Henson built a small model of his aircraft, he could not find anyone ready to put up the cash for a full-size version and rapidly abandoned aerial experimentation for good.

LIGHTER-THAN-AIR FLIGHT

THE BALLOON FLIGHTS OF 1783 began a tradition of lighter-than-air flight that ran parallel with – and for a long time ahead of – experiments with heavier-than-air flight. The drawbacks of balloons were obvious. A huge balloon was needed to carry even a small weight, and then it was only marginally controllable and at the mercy of the winds. Yet serious practical uses were found for balloons in the 19th century: they were employed as observation platforms during the Civil War, and during the Franco-Prussian War they were used to carry messages.

The first controlled powered balloon – a dirigible or airship – was demonstrated by Frenchman Henri Giffard in 1852. Mounting a steam-driven propeller under a cigar-shaped bag filled with coal gas, he flew 17 miles (27km) at around 6mph (10kph). His example inspired other enthusiasts, although they were hindered by the lack of alternatives to the steam engine. In the 1880s electric motors came into vogue, and the *La France*, built by Charles Renard and A.C. Krebs, managed controlled flights at speeds of around 12mph (20kph).

The advent of the internal combustion engine brought a further leap forward. In 1898 Alberto Santos-Dumont, the son of a Brazilian coffee-plantation owner, embarked on a series of highly successful experiments in the skies of Paris, France, where he lived. He became a well-known and popular figure, responding to mishaps, such as crashing on the roof of a hotel, with admirable panache. He built 14 airships in all before transferring his enthusiasm to heavier-than-air flight. Meanwhile, the Germans entered the airship field when Count von Zeppelin flew his first airship LZ 1 in 1900 (see pages 56–57).

BOY FROM BRAZIL

Paris-based Brazilian Alberto Santos-Dumont stands in his balloon basket ready for an ascent. Although an eccentric dilettante, he proved an outstanding pioneer of airship and airplane flight.



BALLOON MEN

The first manned flight was made in a Montgolfier balloon (left, on a cigarette card) on November 21, 1783, but balloon flight had a limited potential. It could not, for example, be used to create a viable transportation system.



TRIP AROUND THE TOWER

In 1901 Alberto Santos-Dumont flew his dirigible from the Parisian suburb of Saint-Cloud around the Eiffel Tower and back in under 30 minutes to win a 100,000-franc prize.

